

OPERATIONAL MODULES

- 🏠

Fisherman Home
- 💬

Ask ORCA

AI Core
- 🗺️

Mission Planner

Trip
- 📖

Tactical Marine Map
- 🔔

Alert Center

Alerts
- 🛡️

Decision Detail

Audit
- 🕒

Mission History

ROLE WORKSPACES

- 🚢

Fisherman View

●
- 🛡️

Coastal Authority
- 🚨

Disaster Management
- 📦

Marine Research
- 🚢

Fleet Operator

🔧 ACTIVE CRAFTWGS84

Matsya Sagar 1

8.5m • Sassoon Docks Terminal

- 👤

Profile & Vessel
- ⚙️

System Settings
- 🔗

Switch Account

🚢 COMMERCIAL MARITIME OPERATOR • MAHARASHTRA — ALIBAUG / MUMBAI SECTOR

Fleet Dispatch & Voyage Route Clearance

Commercial tug, passenger ferry, and survey operations • Scheduled departure envelope monitoring

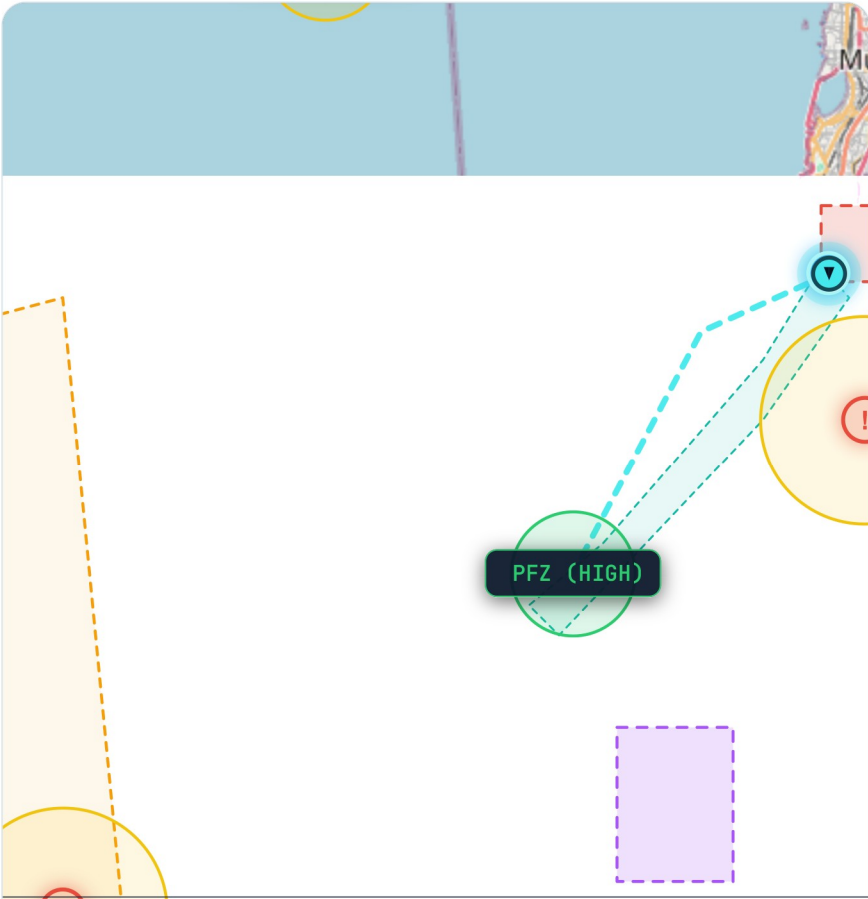
🔗 DISPATCH NEW VOYAGE

OFFLINE FLEET DISPATCH MODE

🚫 Scheduled voyages and departure clearance envelopes reflect cached mission plans. New voyage dispatches require network link or will be queued locally as drafts.

📦 COMMERCIAL ROUTE CORRIDORS

Bathymetric Navigation Corridors



📅 TODAY'S SCHEDULED DEPARTURES

3 Voyages Logged

Sassoon Ferry Express (18m)

CLEARED (PASS)

Passenger / Coastal Transit

Gateway of India & Alibaug Mandwa

06:30 IST → 07:15 IST

Wave: 0.8m (<= 1.8m limit)

Wind: 11 kts (Clear)

Ocean Survey Tug 04 (24m)

CAUTION (AFTERNOON SWELL)

Hydrographic Survey

Mumbai Anchorage & Bombay High Sector

08:00 IST → 16:00 IST

Wave: 1.2m morning & 2.2m afternoon

Wind: 18-22 kts gusts

Alibaug Pelagic

CLEARED (MORNING)

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 DISASTER MANAGEMENT AUTHORITY (NDRF / SDMA) • MAHARASHTRA —
ALIBAUG / MUMBAI SECTOR

Coastal Hazard Exposure & Emergency Coordination

Real-time verified hazard perimeters • Multi-vessel risk exposure • Deterministic
safety directives

 0 Active Hazards

 0 Crafts in Exposure Sector



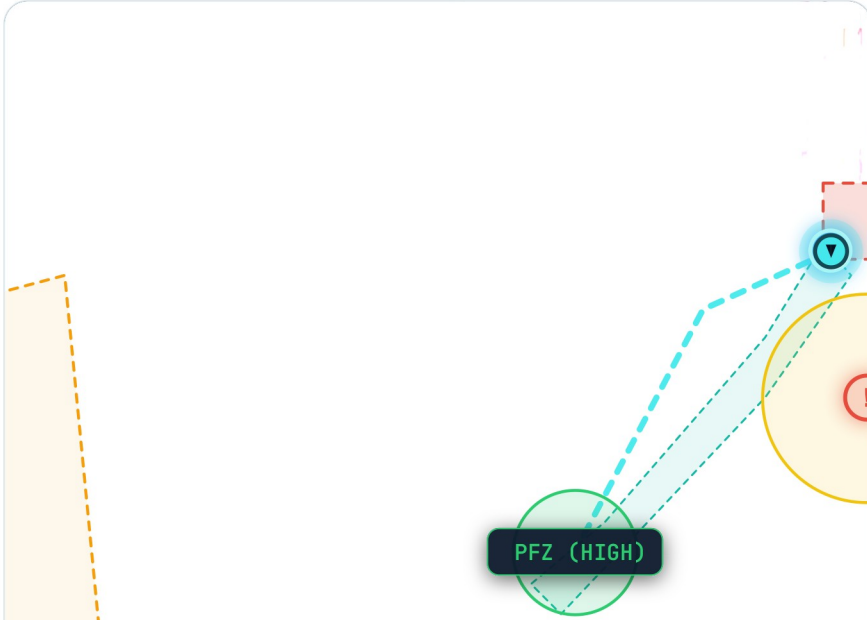
OFFLINE DISASTER RESPONSE MODE GPS: SENSOR_UNAVAILABLE

 Hazard zones and craft exposures are rendered from cached local emergency databases. Live satellite cyclone tracking & wave surge feeds
require active network. Evacuation margins degrade conservatively.

ACTIVE COASTAL HAZARDS

Click a hazard to inspect spatial extent & affected assets

 HAZARD SPATIAL EXTENT & BUFFER PERIMETER PostGIS Deterministic Engine
• WGS84



No active hazard selected.

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AUDIT TRAIL & DECISION REPLAY LEDGER

Mission History & Replay

Immutable record of past mission evaluations, correlated ocean telemetry snapshots, and deterministic audit rules

1
GO

1
CAUTION

1
AVOID

Verdict:

ALL

GO

CAUTION

AVOID

Q

Search missions, vessels, zones...

DECISION ID & TIME	MISSION & CRAFT	TARGET ZONE	VERDICT	PRIMARY RATIONALE	REPLAY
DEC-20260902-01 2026-09-02 08:30 IST	Commercial Fishing (5h) Matsya Sagar 1 (8.5m)	Zone Alpha (Offshore Alibaug) 	CAUTION MODERATE SUPPORT	Significant wave swell height reaches 2.1m post-midday; return window constrained before afternoon sea chop.	Inspect Replay
DEC-20260901-02 2026-09-01 05:15 IST	Artisanal Fishing (3.5h) Matsya Sagar 1 (8.5m)	Murud Ridge (Zone Bravo) 	GO HIGH SUPPORT	Low wave swell (0.9m), calm wind (8 kts), optimal SST chlorophyll thermal front.	Inspect Replay
DEC-20260831-01 2026-08-31 11:00 IST	Deep Sea Pelagic (8h) Samudra Ratna (14m)	North High Deep (Zone Charlie) 	AVOID HIGH SUPPORT	Squall line advisory with wind gusts > 30 kts and swell exceeding 2.8m in outer continental shelf.	Inspect Replay

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EXPLAINABLE MARITIME DECISION AUDIT • MAHARASHTRA – ALIBAUG / MUMBAI SECTOR

Decision Evidence & Reasoning Chain

Transparent 5-level progressive disclosure of physical thresholds, multi-agency evidence, and deterministic rule audits

Ask Follow-up →

DETERMINISTIC OPERATIONAL VERDICT

• MODERATE EVIDENCE SUPPORT

CAUTION

• CAUTION

Trip feasible for early departure at 05:45 IST, but mission duration of 5h borders afternoon wave increase. Conclude operations before midday.

Recommended Departure:

05:45 IST

Safe Return Deadline:

10:45 IST

• Window Duration: ~5h 45m



Deterministic Safety Precedence Invariant: INCOIS PFZ indicates biological fish concentration; it never grants navigational or weather safety clearance. Safety decisions are calculated deterministically on the backend against physical vessel seaworthiness limits and live ocean forecasts.

Level 1: Summary & Window

Level 2: Why (Factors)



Level 3: Multi-Agency Evidence (5)



Level 4: Rule Evaluations (0)

TARGET ZONE & NAVIGATION CORRIDOR

Alibaug Outer Bank (PFZ-MUM-01)

Distance: 18.5 km

Bearing: 245° WSW

TELEMETRY INTEGRITY & FEEDS

5 of 5 Required Feeds Verified

INCOIS OSF (Live) • INCOIS PFZ (Live) • PostGIS GIS (Deterministic) • IMD (Access Pending)



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OPERATIONAL SAFETY & ADVISORY NOTICES • MAHARASHTRA – ALIBAUG / MUMBAI
SECTOR

Active Role: Fisherman ▾



Maritime Hazard & Alert Center

Deterministic safety warnings, PostGIS geofence boundaries, and telemetry
integrity audit

Status:

ACTIVE

ACKNOWLEDGED

RESOLVED

EXPIRED

ALL



Search alerts, rules, areas...

Severity:

ALL

CRITICAL

WARNING

ADVISORY

INFO



No Operational Alerts Found

There are currently no active hazards or warnings matching the selected filters
for Maharashtra — Alibaug / Mumbai Sector.

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MAP LAYERS

Fisherman Authority Disaster Research All

🚢 Active Vessel 1 ✓

📍 User Location Port ✓

🌿 PFZ Fishing Zones 3 ✓

🌊 Weather Risk Overlays Wind/Waves ✓

⚠️ Navigational Hazards 2 ✕

🚫 Geofences & Sanctuaries 2 ✓

🗺️ Recommended Route Active ✓

🛡️ Safe Navigation Corridor Verified ✓

CARTOGRAPHIC LEGEND

● Active Vessel (Matsya Sagar 1) Real-time telemetry

📍 Port / User Location Sassoon Docks

🟢 PFZ Zone (High Potential) Chlorophyll > 1.5 mg/m³

🟡 PFZ Zone (Moderate Potential) Chlorophyll ~ 1.0 mg/m³

— Recommended Route Line Corridor to target PFZ

🚫 Restricted Naval Geofence Strict no-entry zone

🟡 Submerged Shoal / Weather Hazard Clearance required

🛡️ Safe Navigation Corridor Low risk depth envelope

PFZ Vessel Hazard Geofence

PFZ CONTEXT

Zone Alpha (Offshore Alibaug)

STATUS / LEVEL:

HIGH

Optimal pelagic aggregation zone detected via EO Thermal/Chlorophyll Fronts

TARGET COMMERCIAL SPECIES:

Mackerel Sardines Pomfret

POTENTIAL SCORE HIGH

DISTANCE 18.5 km

BEARING 245°

WATER DEPTH 38 m

SST GRADIENT Favorable (27.8°C)

CHLOROPHYLL High (1.82 mg/m³)

SOURCE: DEMO_INCOIS_PFZ_SNAPSHOT

OBSERVED: 2026-09-02T06:00:00Z

VALID UNTIL: 2026-09-03T06:00:00Z

🔍 FOCUS MAP

🗺️ PLAN TRIP



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8.5m • Sassoon Docks Terminal



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MISSION DISPATCH & VOYAGE CLEARANCE • MAHARASHTRA — ALIBAUG / MUMBAI SECTOR

Maritime Mission Planner

Configure voyage route, craft seaworthiness limits, and departure window to evaluate deterministic GIS safety and ocean forecast.



ASK ORCA ABOUT TRIP

1 MISSION IDENTITY & OBJECTIVE

Step 1 of 4

Mission Name

Morning Coastal Fishing Trip

Activity Type



PFZ Target

Commercial Fishing



Ocean Data

Marine Survey



Surveillance

Coastal Patrol

2 ASSIGNED CRAFT & SEAWORTHINESS LIMIT

Step 2 of 4

Select Vessel

Matsya Sagar 1 (8.5m • tradition)

Target Fishing / Patrol Zone

Zone Alpha (Offshore Alibaug) (

Hull Envelope Limit:1.8m max wave

Cruising Speed: 6.5 knots

Water Depth: 38m

SST Gradient: Favorable (27.8°C)

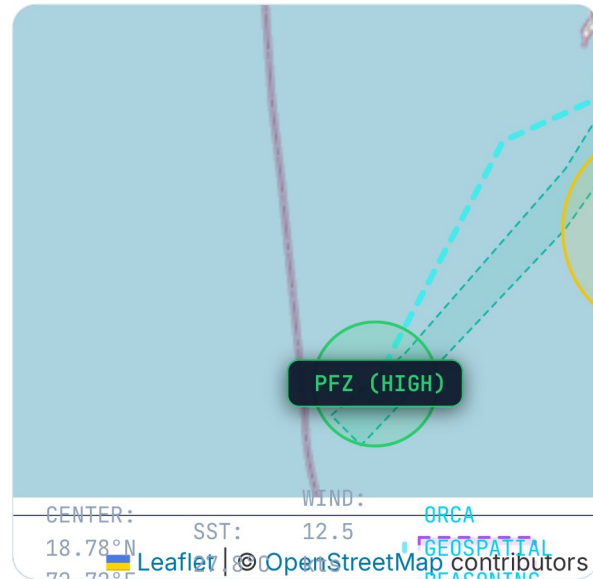
3 DEPARTURE WINDOW & DURATION

Step 3 of 4



VOYAGE CORRIDOR & BOUNDARIES

PostGIS Clearance



- Origin: Sassoon Docks, Mumbai
- Target Zone: Zone Alpha (Offshore Alibaug)

DETERMINISTIC SAFETY CLEARANCE

• CAUTION

CAUTION
RECOMMENDED

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ORCA DECISION ENGINE

SECTOR: MAHARASHTRA — ALIBAUG / MUMBAI SECTOR

🌟

Ask ORCA Intelligence Workspace

Natural-language marine reasoning & What-If scenario intelligence backed by deterministic constraints and live oceanography.

← BACK TO HOME

🌟 INQUIRE MARINE DECISION ENGINE

NATURAL LANGUAGE

🌟 Can I go fishing tomorrow morning for five hours?

📍 EVALUATE

QUICK ACTIONS:

Can I go fishing tomorrow morning for five hours?

Is it safe to transit to Alibaug Outer Bank (PFZ-MUM-01)?

What is the wave swell forecast for afternoon return?

Evaluate patrol corridor clearance near Naval Anchorage

⚙️ WHAT-IF SCENARIO INTELLIGENCE

DETERMINISTIC

Hypothesize timing, vessel, route or wave conditions.

❓ Try: 'What if I leave at 2 PM?' or 'What if waves reach 2.5m?'

⚙️ TEST SCENARIO

PRESETS:

Leave at 2 PM

3-Hour Trip

Use VESSEL-002

Avoid Restricted Zone

Waves 2.5m (Hypothetical)

STRUCTURED SCENARIO CONTROLS:

🕒 DEPARTURE

05:45

06:00

14:00

18:00

🕒 DURATION

5h

3h

5h

8h

🚢 VESSEL

VESSEL-001

VESSEL-001

VESSEL-002

🛡️ OPERATIONAL RATIONALE

Full Audit →

Departure is favorable, but return window approaches worsening midday sea state (> 2.0m swell post-12:00 IST).

Trip feasible for early departure at 05:45 IST, but

✅ RECOMMENDED ACTIONS

- 1 Depart at 05:45 IST for calm morning swell.
- 2 Conclude return by 10:45 IST before afternoon waves.

CAUTION — ADVISORY ACTIVE

EVALUATED AT 05:45 IST



PROCEED WITH CAUTION

Midday Return & Proximity Constraints Active • Target Craft: **Matsya Sagar 1**

🛡️ WHY THIS DECISION? >

🌟 ASK ORCA

OPERATIONAL RATIONALE & SYNTHESIS

Trip feasible for early departure at 05:45 IST, but mission duration of 5h borders afternoon wave increase. Conclude operations before midday.

RECOMMENDED TIMING

Depart 05:45 IST

Conclude Return by 11:30 IST

PFZ OPPORTUNITY (INCOIS)

Zone Alph...

18.5 km • Bearing 245° WSW (High Pelagic)

DETERMINISTIC SAFETY

Clear Navigation Corridor

4.2 km Buffer from Naval Geofence

● All 5 marine observation feeds correlated

🗺️ Inspect on Marine Map →

🗺️ SPATIAL CORRIDOR

Expand Map →



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8.5m • Sassoon Docks Terminal



Profile & Vessel



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MAHARASHTRA — ALIBAUG / MUMBAI SECTOR • ARABIAN SEA

Fisherman Operational Home

PLAN MISSION

Active Vessel: **Matsya Sagar 1** (8.5m • traditional_motorized) • Port: Sassoon Docks, Mumbai



ASK ORCA MARINE DECISION INTELLIGENCE

Correlating Ocean • IMD • PFZ • PostGIS



Can I go fishing tomorrow morning for five hours?

ASK ORCA →

QUICK INQUIRIES:

Can I fish tomorrow morning for 5 hours?

What is the wave swell at Alibaug Outer Bank?

Where is the highest yield INCOIS PFZ front?

What if departure shifts to 2 PM?

CAUTION — ADVISORY ACTIVE



EVALUATED AT 05:45 IST

PROCEED WITH CAUTION

Midday Return & Proximity Constraints Active • Target Craft:
Matsya Sagar 1



WHY THIS DECISION? >



ASK ORCA

OPERATIONAL RATIONALE & SYNTHESIS

Trip feasible for early departure at 05:45 IST, but mission duration of 5h borders afternoon wave increase. Conclude operations before midday.



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Depart 05:45 IST

Conclude Return by 11:30 IST



PFZ OPPORTUNITY
(INCOIS)

Zone Alpha (Alibaug Oute...

18.5 km • Bearing 245° WSW
(High Pelagic)



DETERMINISTIC SAFETY

Clear Navigation Corridor

4.2 km Buffer from Naval
Geofence

